

Adaptive Policy-Based Methods for Informative Power-System Probing

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Abstract—Active probing can reduce uncertainty in power-system dynamics by collecting informative measurements about unknown inertia and frequency-response parameters. In sequential probing, each pulse affects both the current measurement and the physical state carried into later experiments. This paper compares three adaptive policy-based methods—deep adaptive design (DAD), reinforcement-learning-based sequential Bayesian optimal experimental design (RL-sBOED), and Step-DAD—with Random, Fixed, and Myopic baselines on reduced IEEE 9-bus and IEEE 14-bus systems. Across three to five probes, the adaptive policy-based methods achieve higher mean sequential prior-contrastive information scores than the baselines. On IEEE 9-bus, DAD improves over Fixed by 0.09–0.13 nats while requiring only 1.01–1.75 ms per sequence for online decisions on the profiled hardware. Step-DAD provides small additional gains on IEEE 9-bus but at substantially higher online cost, with no consistent improvement on IEEE 14-bus. Overall, the results show that adaptive policy-based probing can improve information acquisition while reducing the online computation required for sequential experiment design.

Index Terms—Power-system identification, active probing, Bayesian experimental design, adaptive policies, expected information gain.

I. INTRODUCTION

Reliable power-system operation depends on knowing how the grid responds to disturbances. Inertia (M) and frequency response (K) are important for short-term frequency dynamics, and their effective values can change as inverter-based resources alter the generation mix [1]. Disturbance-based methods estimate inertia from measured frequency and active-power responses [2]. Active probing takes a different approach: a small test signal is injected into the system and the resulting response is measured. The design of that signal affects the information obtained from the experiment. This has been shown for electromechanical-mode identification [3] and, more recently, for inertia and frequency-response estimation in systems with inverter-based resources [4].

This paper considers a sequence of probing pulses. The pulse amplitude and application bus are fixed, and the duration is selected at each stage. The system is not reset between probes. As a result, each duration changes the current response and also the physical state carried into the next probing

window. The value of a duration therefore depends on the measurements and decisions that came before it. This makes probe selection a sequential design problem, while repeated online search can add substantial computation to the experiment. The setting is related to excitation design in system identification [5], but here the design is updated over a carried-state probing sequence.

Bayesian optimal experimental design (BOED) provides a way to choose probes by maximizing expected information gain about the unknown parameters [6]. A Fixed design can optimize the full sequence before probing begins, but it cannot use measurements collected during the experiment. A Myopic design can react to those measurements, but it optimizes only the next probe and repeats the search online. We refer to DAD [7], RL-sBOED [8], and Step-DAD [9] collectively as *adaptive policy-based methods*. DAD and RL-sBOED learn history-dependent policies offline, whereas Step-DAD starts from a trained DAD policy and further refines it online as measurements become available.

We evaluate these adaptive policy-based methods for continuous pulse-duration selection on reduced IEEE 9-bus and IEEE 14-bus systems. Random, Fixed, and Myopic designs are used as baselines. All six methods use the same physical model, endpoint RoCoF sensor, duration constraints, and probe budget. The comparison focuses on two questions: how much information is acquired over the probing sequence and how much computation is required to choose the probes. We also rescore the IEEE 14-bus histories with larger contrast sets to examine sensitivity to the information estimator. Code and archived results are available online.¹

II. SEQUENTIAL PROBING FORMULATION

We consider T consecutive probing stages. The index $t = 1, \dots, T$ denotes the stage, d is physical time within a stage, and D_t is the pulse duration selected at stage t .

A. Uncertain Parameters and Reduced Swing Dynamics

For a reduced network with m retained machines, the unknown parameter vector is $\theta = (M_1, \dots, M_m, K_1, \dots, K_m)$,

¹https://github.com/alexschmidt123/tpec2027_power_grid

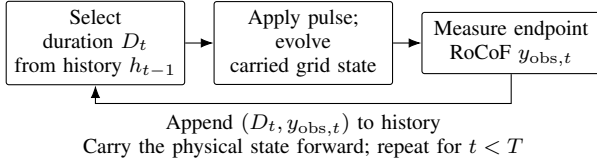


Fig. 1. Sequential probing with state carryover. Adaptive policy-based methods and Myopic use the observed history to select the next duration; Fixed and Random do not use measurement feedback.

where M_i is the inertia coefficient and K_i is the effective frequency-response coefficient for machine i . Let $\bar{M}_i$ and $\bar{K}_i$ denote their respective nominal values. We assume independent uniform priors, $M_i \sim \mathcal{U}(0.7\bar{M}_i, 1.3\bar{M}_i)$ and $K_i \sim \mathcal{U}(0.1\bar{K}_i, \bar{K}_i)$, with joint distribution $p_0(\theta)$. One parameter vector is drawn at the start of each probing sequence and remains fixed throughout that sequence.

Rotor angles δ_i and angular-speed deviations ω_i satisfy

$$\begin{aligned} \dot{\delta}_i &= \omega_i, \\ M_i \dot{\omega}_i &= u_i - \left(c_i + \frac{K_i}{2\pi}\right) \omega_i - \sum_{j=1}^m B_{ij} \sin(\delta_i - \delta_j), \end{aligned} \quad (1)$$

Here, u_i is the probing input, c_i is the known damping coefficient, and B_{ij} is the network coupling. Dots denote derivatives with respect to d , and the frequency deviation is $\Delta f_i = \omega_i/(2\pi)$ Hz. The coefficient K_i represents an effective power response to frequency deviation; turbine-governor dynamics are not modeled separately. We use a 100-MVA, 60-Hz base. The initial rotor angles, speed deviations, and net mechanical injections are zero, so each sequence starts from an unforced equilibrium.

The reduced IEEE 9-bus model retains buses (1, 2, 3), and the IEEE 14-bus model retains buses (1, 2, 3, 6, 8). A lossless network Laplacian $\mathbf{L}$ is formed from the MATPOWER test-case reactances [10]. Algebraic buses are removed by Kron reduction, $\mathbf{L}_{\text{red}} = \mathbf{L}_{gg} - \mathbf{L}_{gl}\mathbf{L}_{ll}^{-1}\mathbf{L}_{lg}$, with $B_{ij} = -(\mathbf{L}_{\text{red}})_{ij}$ for $i \neq j$. Resistance, voltage dynamics, and transformer tap effects are omitted.

Nominal inertia is computed as $\bar{M}_i = 2H_i/(2\pi f_0)$ with $f_0 = 60$ Hz. For the IEEE 9-bus system, the inertia constants $H = (23.64, 6.40, 3.01)$ s are taken from [11]. For IEEE 14-bus, we use $H = (4, 6.5, 5, 5, 5)$ s as a benchmark setting. The nominal response coefficient is $\bar{K}_i = K_{\text{tot}} w_i$, where $K_{\text{tot}} = 0.5$ pu/Hz. The weights are $w = (0.5875, 0.2625, 0.15)$ for IEEE 9-bus and $w = (8, 13, 10, 10, 10)/51$ for IEEE 14-bus. Damping is set to $c_i = (0.1 \text{ s}^{-1})\bar{M}_i$. Table I lists the resulting values. The IEEE 14-bus inertia, response, damping, and prior ranges are benchmark assumptions and should not be interpreted as validated operating parameters.

B. Pulse Design, State Carry, and Sensing

Each probe is described by $\xi_t = (A_t, B_t, D_t)$, where A_t is the amplitude, B_t is the application bus, and D_t is the duration.

TABLE I
NOMINAL PARAMETER SCALES AND FIXED DAMPING

Network	Bus	$\bar{M}_i$	$\bar{K}_i$	c_i
IEEE 9-bus	1	0.1254141	0.2937500	0.0125414
	2	0.0339531	0.1312500	0.0033953
	3	0.0159685	0.0750000	0.0015969
IEEE 14-bus	1	0.0212207	0.0784314	0.0021221
	2	0.0344836	0.1274510	0.0034484
	3, 6, 8	0.0265257	0.0980392	0.0026526

We fix $A_t = 0.05$ pu and $B_t = 1$ and optimize only D_t . The probe has a raised-cosine shape,

$$u_{B_t,t}(d) = \frac{A_t}{2} [1 - \cos(2\pi d/D_t)], \quad 0 \leq d \leq D_t, \quad (2)$$

and is zero after D_t and at all other buses. Every stage uses a fixed $D_{\text{obs}} = 3.5$ s recording window. The terminal rotor angles and speed deviations are carried directly into the next stage. Thus, the response at stage t depends on the durations applied earlier in the same sequence.

The noise-free scalar observation is the signed endpoint rate of change of frequency (RoCoF) at bus 1:

$$y_t(\theta, \xi_{1:t}) = \frac{\Delta f_{1,t}(D_{\text{obs}}) - \Delta f_{1,t}(D_{\text{obs}} - \Delta d_{\text{samp}})}{\Delta d_{\text{samp}}}, \quad (3)$$

Here, $\xi_{1:t} = (\xi_1, \dots, \xi_t)$ denotes the sequence of applied probes. The frequency deviation at bus 1 during stage t is obtained from the angular-speed deviation in (1) as $\Delta f_{1,t}(d) = \omega_{1,t}(d)/(2\pi)$. The endpoint RoCoF is calculated from the frequency difference over the final sampling interval, $\Delta d_{\text{samp}} = 0.025$ s. Measurement noise is modeled as

$$y_{\text{obs},t} = y_t(\theta, \xi_{1:t}) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (4)$$

with independent Gaussian noise and $\sigma = 0.005$ Hz/s. The sensor therefore reports signed RoCoF at the end of the recording window, not the initial inertial RoCoF or the maximum absolute RoCoF during the stage. The next duration is selected after this measurement is available. Decision latency is not modeled in the physical simulation.

Durations satisfy the common constraints

$$\begin{aligned} 0.2 &\leq D_t \leq 3, \quad t = 1, \dots, T, \\ D_t - D_{t-1} &\geq \Delta D_{\min}, \quad t = 2, \dots, T, \end{aligned} \quad (5)$$

where $\Delta D_{\min} = 0.01$ s. At each stage, the feasible interval is shortened as needed so that the remaining probes can still satisfy the increasing-duration constraint. The same continuous action mapping is used for all methods. This ordering rule is a benchmark restriction rather than a requirement of the swing equations. Since the probing amplitude is fixed, different pulse durations lead to different amounts of injected energy. Therefore, the methods are compared using the same number of probes rather than the same energy budget.

C. History-Dependent Decisions and Bayesian Updating

After stage t , the observed history is $h_t = ((\xi_k, y_{\text{obs},k}))_{k=1}^t$, with $h_0 = \emptyset$. A policy $\pi \in \Pi$ selects the next probe from this history only; it does not observe the true parameter vector or the true internal state. For any candidate θ , the simulator predicts the response to the applied duration sequence and defines the Gaussian likelihood in (4). Bayes' rule gives

$$p(\theta | h_t) = \frac{p(y_{\text{obs},t} | \theta, h_{t-1}, \xi_t) p(\theta | h_{t-1})}{p(y_{\text{obs},t} | h_{t-1}, \xi_t)}. \quad (6)$$

with $p(\theta | h_0) = p_0(\theta)$. Each candidate parameter vector carries its own predicted state through the observed duration sequence. Since the applied action is already determined by the observed history, the posterior update does not require a policy-action probability term.

D. Full-Sequence Information Objective

Expected information gain (EIG) is

$$\mathcal{I}_T(\pi) = \mathbb{E}_{h_T|\pi} [D_{\text{KL}}(p(\theta | h_T) || p_0(\theta))], \quad (7)$$

where D_{KL} is the Kullback–Leibler divergence. The expectation is taken over parameter draws, measurement noise, and any policy randomness. We seek a policy that solves

$$\pi^* \in \arg \max_{\pi \in \Pi} \mathcal{I}_T(\pi). \quad (8)$$

Information is measured in nats. Fixed and the three adaptive policy-based methods are evaluated with the full-sequence objective. Myopic approximately maximizes one-step conditional information gain, while Random does not optimize an information objective.

Direct EIG evaluation requires integrating the nonlinear likelihood over the prior. We instead use the sequential prior-contrastive (sPCE) lower bound [7]. One prior draw, $\theta^{(0)}$, generates the observed sequence, and L additional prior draws serve as contrasts. For a candidate parameter vector, the likelihood of the observed history is

$$\mathcal{L}_t(\theta) = \prod_{k=1}^t p(y_{\text{obs},k} | \theta, h_{k-1}, \xi_k). \quad (9)$$

The sampled score is

$$\hat{I}_t = \log \frac{\mathcal{L}_t(\theta^{(0)})}{\frac{1}{L+1} \sum_{\ell=0}^L \mathcal{L}_t(\theta^{(\ell)})}. \quad (10)$$

Likelihoods are evaluated in the log domain. The mean terminal score estimates the sPCE lower bound rather than EIG itself. A sampled score can be negative and is bounded above by $\log(L+1)$; for $L = 128$, the ceiling is 4.8598 nats. We therefore call the reported values *information scores* and later check their sensitivity to larger contrast sets. The generating parameters are used by the simulator and scorer only and are never given to a design method.

III. DESIGN METHODS

All six methods use the same simulator, observation model, and duration constraints. The main differences are whether measurements are used to adapt later probes, whether the objective covers one step or the full sequence, and whether optimization is performed offline or during probing.

A. Nonadaptive Baselines: Random and Fixed

Random samples one unit action at each stage and maps it to the remaining feasible duration interval. It uses no measurement feedback and requires no offline training. Because the feasible interval changes after each decision, the resulting complete sequences are not uniformly distributed.

Fixed optimizes the full duration sequence before test measurements are observed, following the nonadaptive formulation in [6, Sec. 2.4]. It uses terminal sPCE as the objective but does not adapt during probing. Several interior initializations are compared first, after which the best initialization is optimized with pathwise gradients. A separate Fixed sequence is obtained for each training seed.

B. Online One-Step Baseline: Myopic

Myopic uses the current history to choose the next duration by approximately maximizing one-step conditional information gain [6, Sec. 2.4]. The posterior is represented by 128 weighted particles, each carrying its own physical state. At every decision, three rounds of 16 candidate durations are searched, using 32 simulated observations per candidate. No offline training is required, but posterior updating and candidate search are repeated during probing. The selected action is therefore an approximate myopic optimum.

C. Adaptive Policy-Based Methods

The adaptive policy-based methods use the observed history to select later probes. Their policy input contains the stage index, previous durations, scaled observations, and masks for unused history slots. Past probes remain ordered because the physical state is carried across stages. DAD and RL-sBOED learn the policy offline and select a duration at test time with a policy forward pass followed by the common feasible-duration mapping. Step-DAD uses the same history representation but adds online refinement.

DAD [7] uses two 64-unit tanh hidden layers, a trainable stage bias, and a sigmoid action map. Training uses pathwise gradients through the observations and carried states to maximize terminal sPCE. Before training, a random initialization is compared with an initialization based on the optimized Fixed sequence. The reported DAD preparation time includes this Fixed stage.

RL-sBOED [8] is implemented with a REDQ-style actor-critic using ten critics and five replay updates per trajectory batch. The actor has two 128-unit ReLU hidden layers. Its stage reward is $R_t = \hat{I}_t - \hat{I}_{t-1}$ with $\hat{I}_0 = 0$, so the undiscounted rewards sum to terminal sPCE. Training uses entropy regularization with target entropy -1 in normalized action coordinates, and evaluation uses the deterministic actor

output. Unlike DAD, RL-sBOED does not require pathwise differentiation through the swing simulator.

Step-DAD [9] starts from the selected DAD checkpoint. The first duration comes directly from the pretrained policy. After each subsequent measurement, a 128-particle posterior is used to simulate the remaining sequence, and the policy receives four Adam updates with batch size 32 and learning rate 10^{-3} . A conditional validation batch determines whether the updated weights are retained. Accepted updates carry forward within the current probing sequence, while each new sequence starts from the original DAD checkpoint. Step-DAD is therefore adaptive at the policy level but retains substantial online computation.

IV. EXPERIMENTAL PROTOCOL

A. Simulation, Training, and Checkpoint Selection

Experiments use $T = 3, 4, 5$ probes, with one scalar RoCoF measurement per stage. A separate policy is trained for each network and horizon. The swing equations are integrated with fourth-order Runge–Kutta using a 1/640-s time step. DAD and Step-DAD apply the chain rule through numerical Jacobians of the state and duration. State perturbations are $10^{-5} \max(1, |x_j|)$ and duration perturbations are 10^{-5} s, clipped to the feasible range. The resulting policy and refinement gradients are also checked against end-to-end finite differences.

Fixed, DAD, and RL-sBOED use training seeds 101, 202, and 303. Each run performs 2000 trajectory updates with batch size 32. Fixed and DAD use Adam with learning rate 10^{-3} ; the RL actor and critics use 3×10^{-4} . Checkpoints are evaluated every 100 updates on 128 validation systems. Training, validation, and the initial evaluation use $L = 128$ contrasts. The shared update count provides a common protocol, but it should not be interpreted as equal optimization effort or proof of convergence.

B. Paired Evaluation and Statistical Reporting

Testing uses seeds 1001, 1002, and 1003, with 512 probing sequences per seed. Each fitted policy is therefore evaluated on 1536 sequences. Parameter and measurement-noise draws are paired across methods when the protocol allows; planner randomness is generated separately. Training, validation, and test seeds are distinct.

For Fixed and the adaptive policy-based methods, each reported value is the mean and sample standard deviation (SD) across three training-seed means, where each seed mean already averages all test sequences. Random and Myopic have no training seed, so their SDs are computed across the three test-seed means. The SDs therefore summarize seed-to-seed variation, not episode-level variability or uncertainty in paired method differences. We use these comparisons descriptively and do not make significance claims from the separate SDs.

TABLE II
IEEE 9-BUS SPCE SCORES ($L = 128$; NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.4321 ± 0.0310	0.4924 ± 0.0342	0.5392 ± 0.0471
Fixed	2.0112 ± 0.0116	2.3192 ± 0.0004	2.5066 ± 0.0013
Myopic	1.9736 ± 0.0438	2.1707 ± 0.0366	2.2407 ± 0.0547
DAD	2.1366 ± 0.0161	2.4095 ± 0.0080	2.6351 ± 0.0179
RL-sBOED	2.1257 ± 0.0027	2.3963 ± 0.0333	2.5307 ± 0.0114
Step-DAD	2.1471 ± 0.0110	2.4382 ± 0.0048	2.6621 ± 0.0145

TABLE III
IEEE 14-BUS SPCE SCORES ($L = 128$; NATS)

Method	$T = 3$	$T = 4$	$T = 5$
Random	0.7541 ± 0.0047	0.8360 ± 0.0297	0.8464 ± 0.0252
Fixed	4.2623 ± 0.0402	4.4946 ± 0.0345	4.6624 ± 0.0083
Myopic	4.2907 ± 0.0368	4.3990 ± 0.0177	4.4641 ± 0.0280
DAD	4.7576 ± 0.0362	4.8325 ± 0.0047	4.8443 ± 0.0026
RL-sBOED	4.7445 ± 0.0022	4.8275 ± 0.0112	4.8383 ± 0.0108
Step-DAD	4.7527 ± 0.0275	4.8297 ± 0.0067	4.8446 ± 0.0042

V. RESULTS AND DISCUSSION

A. Information Scores at Matched Probe Counts

Tables II and III report the $L = 128$ information scores; the highest mean score in each horizon is shown in bold. On both networks and at every tested horizon, the adaptive policy-based methods have higher mean scores than Random, Fixed, and Myopic. For IEEE 9-bus, DAD exceeds Fixed by 0.1253, 0.0903, and 0.1286 nats at $T = 3, 4, 5$. Step-DAD adds 0.0105, 0.0287, and 0.0269 nats over DAD. The adaptive policy-based methods achieve higher information scores than the baselines under the tested settings. However, because the methods differ in their design strategies and optimization procedures, these results do not establish that adaptation alone accounts for the improvement.

On IEEE 9-bus, Fixed also scores above Myopic at all three horizons. Feedback alone is therefore not enough to guarantee a better design when the online search is one-step and approximate. The IEEE 14-bus results require additional care because the adaptive policy-based scores lie close to the 4.8598-nat ceiling of the $L = 128$ estimator. Step-DAD differs from DAD by only -0.0049 , -0.0028 , and $+0.0003$ nats for $T = 3, 4, 5$. These small differences do not establish which method has higher true expected information gain.

B. Sensitivity to the Information Estimator

We therefore rescore all 64 512 archived IEEE 14-bus histories with $L = 10^4$ and 10^5 contrasts. No policy is retrained or rerun. The generating parameters, durations, measurements, and histories remain fixed, so only the information estimator changes. Matching test-seed and sequence pairs use the same contrast streams across methods, and the 10^4 set is a prefix of the 10^5 set. Table IV reports both rescoring settings using the same seed aggregation as the main evaluation. The highest mean score in each horizon is shown in bold.

TABLE IV
IEEE 14-BUS RESCORED SPCE SCORES (NATS)

(a) $L = 10^4$			
Method	$T = 3$	$T = 4$	$T = 5$
Random	0.8449 ± 0.0121	0.9552 ± 0.0162	0.9991 ± 0.0388
Fixed	5.2956 ± 0.0802	6.0609 ± 0.1246	6.7906 ± 0.0167
Myopic	5.8136 ± 0.0514	6.4715 ± 0.0127	6.8192 ± 0.0500
DAD	7.1062 ± 0.2327	8.2061 ± 0.0286	8.6662 ± 0.0556
RL-sBOED	7.0793 ± 0.0567	8.1296 ± 0.1355	8.5852 ± 0.2660
Step-DAD	7.1008 ± 0.2309	8.1992 ± 0.0191	8.6583 ± 0.0550
(b) $L = 10^5$			
Method	$T = 3$	$T = 4$	$T = 5$
Random	0.8476 ± 0.0136	0.9645 ± 0.0150	1.0075 ± 0.0333
Fixed	5.3472 ± 0.0837	6.1694 ± 0.1401	7.0488 ± 0.0272
Myopic	5.9348 ± 0.0697	6.7628 ± 0.0277	7.2307 ± 0.0540
DAD	7.3126 ± 0.2674	8.8217 ± 0.0707	9.6931 ± 0.1308
RL-sBOED	7.2873 ± 0.0578	8.7141 ± 0.2070	9.5528 ± 0.4753
Step-DAD	7.3064 ± 0.2650	8.8135 ± 0.0578	9.6947 ± 0.1220

With the larger contrast sets, the score ceilings rise to 9.2104 and 11.5129 nats. The broad result is unchanged: all three adaptive policy-based methods remain above Random, Fixed, and Myopic in mean score. However, Myopic now exceeds Fixed at every horizon, reversing their $L = 128$ ordering at $T = 4$ and $T = 5$. The reversal shows that finite contrast count can affect comparisons between closely performing methods.

At $L = 10^5$, the differences of Step-DAD from DAD are -0.0062 , -0.0082 , and $+0.0016$ nats for $T = 3, 4, 5$, so online refinement still gives no consistent mean advantage on IEEE 14-bus. DAD's $T = 5$ score also changes from 8.666 to 9.693 nats between $L = 10^4$ and 10^5 . Increasing the number of contrasts preserves the advantage of the adaptive policy-based methods over the baselines. However, the scores remain sensitive to the contrast count, so the results do not establish a reliable EIG ranking among the adaptive policy-based methods.

C. Selected Probe Durations

Figure 2 shows the selected durations for IEEE 14-bus at $T = 5$. Values are pooled over test sequences and, for trained methods, over the three training seeds. Each fitted method contributes 4608 histories; Random and Myopic contribute 1536 each. Markers denote medians and bars denote interquartile ranges.

The final-stage medians are 0.369s for DAD, 0.376s for Step-DAD, 1.114s for Fixed, 2.283s for Myopic, and 2.979s for Random. Random shifts toward the upper bound at later stages because each new duration is sampled from the remaining feasible interval. RL-sBOED shows stronger variation across training seeds, with final-stage medians of 2.585s, 0.381s, and 0.324s for seeds 101, 202, and 303. This seed variation accounts for much of its wide pooled range. The figure compares the probing strategies selected by different methods, rather than their information scores. Although DAD and Step-DAD show similar duration distributions, the plotted medians do not capture how individual policies respond to pre-

TABLE V
COMPUTATIONAL COSTS ON A100-EQUIPPED SYSTEMS

Method	IEEE 9-bus			IEEE 14-bus		
	$T = 3$	$T = 4$	$T = 5$	$T = 3$	$T = 4$	$T = 5$
Offline preparation (hours)						
Random	0	0	0	0	0	0
Fixed	0.554	0.802	1.063	1.880	2.750	3.720
Myopic	0	0	0	0	0	0
DAD	1.116	1.608	2.139	3.810	5.457	7.418
RL-sBOED	0.100	0.117	0.129	0.163	0.205	0.249
Step-DAD	1.116	1.608	2.139	3.810	5.457	7.418
Online decisions (ms per sequence)						
Random	0.167	0.229	0.303	0.158	0.207	0.285
Fixed	0.849	1.135	1.472	1.141	1.906	1.952
Myopic	4418	5952	7514	9998	13433	16918
DAD	1.012	1.345	1.745	1.440	1.513	1.733
RL-sBOED	1.147	1.781	7.692	1.413	1.550	1.905
Step-DAD	3083	7689	14346	9245	23320	45089

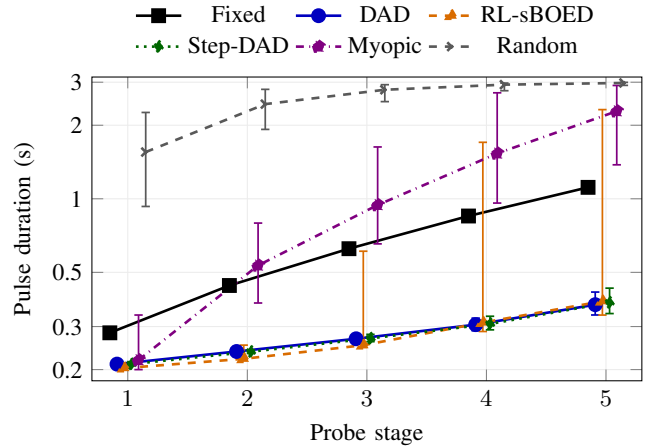


Fig. 2. Selected pulse durations for IEEE 14-bus at $T = 5$. Markers show pooled medians and bars show the 25th–75th percentiles. The vertical axis is logarithmic.

vious observations. The sPCE score depends on the complete sequence of probes and measurements, not only on the selected durations. Furthermore, changing the number of contrasts L affects the estimated information scores but does not change the durations in the saved probing histories.

D. Offline Preparation and Online Decision Cost

Table V reports offline preparation and online decision time on NVIDIA A100 PCIe 40-GB systems using training seed 101 and test seed 1001. Online values are the mean total decision time over 512 test sequences. For IEEE 9-bus, one warm-up sequence per method is excluded. Offline values are single-run measurements and include initialization, optimization, and checkpoint validation. Because DAD initialization is compared against an optimized Fixed design, the DAD and Step-DAD preparation times include the Fixed stage.

The online timer includes action selection and, for Myopic and Step-DAD, posterior updating and refinement. It excludes

propagation of the true test system, measurement acquisition, and terminal information scoring. Physical simulation runs on CUDA while policy and planning code also use the host, so the reported values are implementation-level decision times rather than pure neural-network inference times. Timing runs use the same $L = 128$ scoring setting as the main experiments. Myopic uses 128 planning particles and 32 simulated observations per candidate duration. The lowest cost in each network–horizon column is shown in bold; ties are both bolded.

The main computational difference appears in the online decision times. DAD and RL-sBOED select durations with trained policies, whereas Myopic repeatedly simulates candidate probes. On IEEE 9-bus, DAD requires 1.012–1.745ms per sequence for $T = 3$ –5, compared with 4.418–7.514s for Myopic. DAD pays for this reduction with 1.116–2.139h of offline preparation, which is performed before the test sequences.

Step-DAD makes a different tradeoff. At IEEE 9-bus $T = 5$, its mean score is 0.0269 nats above DAD, but online decision time rises from 1.745ms to 14.346s per sequence. At IEEE 14-bus $T = 5$, Step-DAD requires 45.089s and still shows no consistent score improvement over DAD across horizons. Under this implementation, the small benefit of online refinement comes with a large decision-time cost. These are average implementation times, not worst-case latency guarantees.

For a dense reduced system with m machines, one RK4 simulation over N integration steps has a cost of $C_s = O(Nm^2)$. Random and Fixed require little online computation because neither performs online design optimization. DAD and RL-sBOED select probe durations through policy forward passes, with a total policy-inference cost of $O(TC_\pi)$, where C_π is the cost of one forward pass. In contrast, Myopic repeatedly evaluates candidate durations using posterior particles and simulated observations. Step-DAD requires additional computation for posterior updating and online policy refinement. These differences account for the substantial variation in online decision times reported in Table V.

E. Scope and Limitations

The study uses reduced lossless models, fixed probe and sensor locations, the stated priors, and an imposed increasing-duration constraint. Equal probe counts do not imply equal injected energy. Training and testing also use the same prior, so prior mismatch, sensor bias, and unmodeled dynamics are outside the present evaluation. Myopic and Step-DAD use finite particles and limited online optimization and should not be interpreted as exact optima. Most importantly, sPCE is a lower bound on information about the joint parameter vector. It does not directly measure parameter-estimation error, frequency-security improvement, or closed-loop control performance. No explicit safety constraint is included, and the physical simulation assumes zero decision latency. The timing results therefore characterize algorithmic decision cost rather than complete real-time deployment feasibility.

VI. CONCLUSION

This paper compared adaptive policy-based methods with Random, Fixed, and Myopic designs for sequential active probing with state carryover. On the reduced IEEE 9-bus and IEEE 14-bus systems, DAD, RL-sBOED, and Step-DAD achieve higher mean sPCE information scores across three to five probes. DAD and RL-sBOED perform most design optimization offline and keep online decision cost low relative to Myopic. Step-DAD provides small additional gains on IEEE 9-bus but requires much more online computation and gives no consistent improvement on IEEE 14-bus. Larger-contrast rescoring preserves the overall advantage of the adaptive policy-based methods while showing that close rankings remain estimator-sensitive. Future work should examine richer grid dynamics, model mismatch, practical sensing and latency, and safety-aware probing constraints.

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